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Q1. Smart Hospital Patient Registration System

Score: 0.00 TC: 0/3 passed

CODE

```
import java.util.*;

class Main
{
    public static String readInput(Scanner sc)
    {
        String s="";

        while(sc.hasNextLine())
        {
            s=sc.nextLine().trim();

            if(!s.isEmpty())
                return s;
        }

        return s;
    }

    public static String properCase(String str)
    {
        str=str.trim().toLowerCase();

        String words[]=str.split("\\s+");

        String ans="";

        for(int i=0;i<words.length;i++)
        {
            ans+=Character.toUpperCase(words[i].charAt(0));

            if(words[i].length()>1)
                ans+=words[i].substring(1);

            if(i!=words.length-1)
                ans+=" ";
        }

        return ans;
    }

    public static void main(String args[])
    {
        Scanner sc=new Scanner(System.in);

        ArrayList<Patient> patientList=new ArrayList<Patient>();
        LinkedHashMap<String,Patient> patientMap=new LinkedHashMap<String,Patient>();

        while(true)
        {
            int choice=Integer.parseInt(readInput(sc));

            if(choice==1)
            {
                String id=readInput(sc);
                String name=properCase(readInput(sc));
                String mobile=readInput(sc);
                String department=readInput(sc).toUpperCase();
                String doctor=readInput(sc);

                if(patientMap.containsKey(id))
```

```

        {
            System.out.println("Duplicate Patient ID");
            System.out.println();
            continue;
        }

        if(!mobile.matches("\\d{10}"))
        {
            System.out.println("Invalid Mobile Number");
            System.out.println();
            continue;
        }

        Patient p=new Patient(id,name,mobile,department,doctor);

        patientList.add(p);
        patientMap.put(id,p);

        System.out.println("Patient Registered Successfully.");
        System.out.println();
    }

    else if(choice==2)
    {
        String search=readInput(sc);

        if(search.equalsIgnoreCase("Search Patient:"))
        {
            search=readInput(sc);
        }

        if(patientMap.containsKey(search))
        {
            Patient p=patientMap.get(search);

            System.out.println("Patient Details");
            System.out.println();

            System.out.println("ID : "+p.getId());
            System.out.println();
            System.out.println("Name : "+p.getName());
            System.out.println();
            System.out.println("Mobile : "+p.getMobile());
            System.out.println();
            System.out.println("Department : "+p.getDepartment());
            System.out.println();
            System.out.println("Doctor : "+p.getDoctor());
        }
        else
        {
            System.out.println("Patient Not Found");
        }
    }

    else if(choice==3)
    {
        System.out.println("-----");
        System.out.println("ID Name Department");
        System.out.println("-----");

        for(Patient p:patientList)
        {
            System.out.println(p.getId()+" "+p.getName()+" "+p.getDepartment());
        }
    }

```

```
        System.out.println();
    }

    else if(choice==4)
    {
        LinkedHashMap<String,Integer> departmentCount=new LinkedHashMap<String,Integer>();

        for(Patient p:patientList)
        {
            String dept=p.getDepartment();

            if(departmentCount.containsKey(dept))
            {
                departmentCount.put(dept,departmentCount.get(dept)+1);
            }
            else
            {
                departmentCount.put(dept,1);
            }
        }

        System.out.println("Department Count");
        System.out.println();

        for(String dept:departmentCount.keySet())
        {
            System.out.println(dept+" : "+departmentCount.get(dept));
            System.out.println();
        }
    }

    else if(choice==5)
    {
        break;
    }
}
}
```

```
class Patient
{
    private String id;
    private String name;
    private String mobile;
    private String department;
    private String doctor;

    public Patient()
    {

    }

    public Patient(String id,String name,String mobile,String department,String doctor)
    {
        this.id=id;
        this.name=name;
        this.mobile=mobile;
        this.department=department;
        this.doctor=doctor;
    }

    public String getId()
    {
        return id;
    }
}
```

```
public String getName()
{
    return name;
}

public String getMobile()
{
    return mobile;
}

public String getDepartment()
{
    return department;
}

public String getDoctor()
{
    return doctor;
}

public void setId(String id)
{
    this.id=id;
}

public void setName(String name)
{
    this.name=name;
}

public void setMobile(String mobile)
{
    this.mobile=mobile;
}

public void setDepartment(String department)
{
    this.department=department;
}

public void setDoctor(String doctor)
{
    this.doctor=doctor;
}
}
```

INPUT

```
1
P101
Rahul Kumar
9876543210
Cardiology
Dr. Arun
1
P102
Priya
9876543211
Ortho
Dr. Meena
3
4
2
P101
5
```

OUTPUT

```
Patient Registered Successfully.
Patient Registered Successfully.
```

```
-----
ID Name Department
-----
```

P101 Rahul Kumar CARDIOLOGY
P102 Priya ORTHO

Department Count

CARDIOLOGY : 1

ORTHO : 1

Patient Details

ID : P101

Name : Rahul Kumar

Mobile : 9876543210

Department : CARDIOLOGY

Doctor : Dr. Arun

Q2. Online Library Book Management System**CODE**

```
import java.util.*;

class Main
{
    public static String readInput(Scanner sc)
    {
        String s="";
        while(sc.hasNextLine())
        {
            s=sc.nextLine().trim();
            if(!s.isEmpty())
                return s;
        }
        return s;
    }

    public static String titleCase(String str)
    {
        str=str.trim().toLowerCase();
        String words[]=str.split("\\s+");
        String ans="";

        for(int i=0;i<words.length;i++)
        {
            ans+=Character.toUpperCase(words[i].charAt(0));

            if(words[i].length()>1)
                ans+=words[i].substring(1);

            if(i!=words.length-1)
                ans+=" ";
        }
        return ans;
    }

    public static void main(String args[])
    {
        Scanner sc=new Scanner(System.in);

        ArrayList<Book> bookList=new ArrayList<Book>();
        LinkedHashMap<String,Book> bookMap=new LinkedHashMap<String,Book>();
        LinkedHashSet<String> authorSet=new LinkedHashSet<String>();

        while(true)
        {
            int choice=Integer.parseInt(readInput(sc));

            if(choice==1)
            {
                String id=readInput(sc);
                String title=titleCase(readInput(sc));
                String author=readInput(sc).trim();
                String category=readInput(sc).toUpperCase();
                String publisher=readInput(sc).trim();

                if(bookMap.containsKey(id))
                {
                    System.out.println("Duplicate Book ID");
                    System.out.println();
                    continue;
                }
            }
        }
    }
}
```

```

        if(publisher.isEmpty())
        {
            System.out.println("Invalid Publisher");
            System.out.println();
            continue;
        }

        Book b=new Book(id,title,author,category,publisher);

        bookList.add(b);
        bookMap.put(id,b);
        authorSet.add(author);

        System.out.println("Book Added Successfully.");
        System.out.println();
    }

    else if(choice==2)
    {
        String search=readInput(sc);

        if(bookMap.containsKey(search))
        {
            Book b=bookMap.get(search);

            System.out.println("Book Details");
            System.out.println();

            System.out.println("Book ID : "+b.getId());
            System.out.println();

            System.out.println("Title : "+b.getTitle());
            System.out.println();

            System.out.println("Author : "+b.getAuthor());
            System.out.println();

            System.out.println("Category : "+b.getCategory());
            System.out.println();

            System.out.println("Publisher : "+b.getPublisher());
        }
        else
        {
            System.out.println("Book Not Available");
        }
    }

    else if(choice==3)
    {
        System.out.println("-----");
        System.out.println("Book ID Title Author");
        System.out.println("-----");

        for(Book b:bookList)
        {
            System.out.println(b.getId()+" "+b.getTitle()+" "+b.getAuthor());
        }

        System.out.println();

        else if(choice==4)
    {
        System.out.println("Unique Authors");
        System.out.println();
    }
}

```



```

        for(String author : authorSet)
        {
            System.out.println(author);
        }

        System.out.println();
    }

    else if(choice==5)
    {
        LinkedHashMap<String,Integer> countMap=new LinkedHashMap<String,Integer>();

        for(Book b:bookList)
        {
            String category=b.getCategory();

            if(countMap.containsKey(category))
            {
                countMap.put(category,countMap.get(category)+1);
            }
            else
            {
                countMap.put(category,1);
            }
        }

        System.out.println("Category-wise Count");
        System.out.println();

        for(Map.Entry<String,Integer> e:countMap.entrySet())
        {
            System.out.println(e.getKey()+" : "+e.getValue());
        }

        System.out.println();
    }

    else if(choice==6)
    {
        break;
    }
}

class Book
{
    private String id;
    private String title;
    private String author;
    private String category;
    private String publisher;

    Book(String id,String title,String author,String category,String publisher)
    {
        this.id=id;
        this.title=title;
        this.author=author;
        this.category=category;
        this.publisher=publisher;
    }

    public String getId()
    {

```

```
        return id;
    }

    public String getTitle()
    {
        return title;
    }

    public String getAuthor()
    {
        return author;
    }

    public String getCategory()
    {
        return category;
    }

    public String getPublisher()
    {
        return publisher;
    }
}
```

INPUT

```
1
B101
Java programming
Herbert Schildt
Programming
McGraw Hill
1
B102
Data structures
Mark Allen
Computer science
Pearson
3
4
5
2
B101
6
```

OUTPUT

(no output)